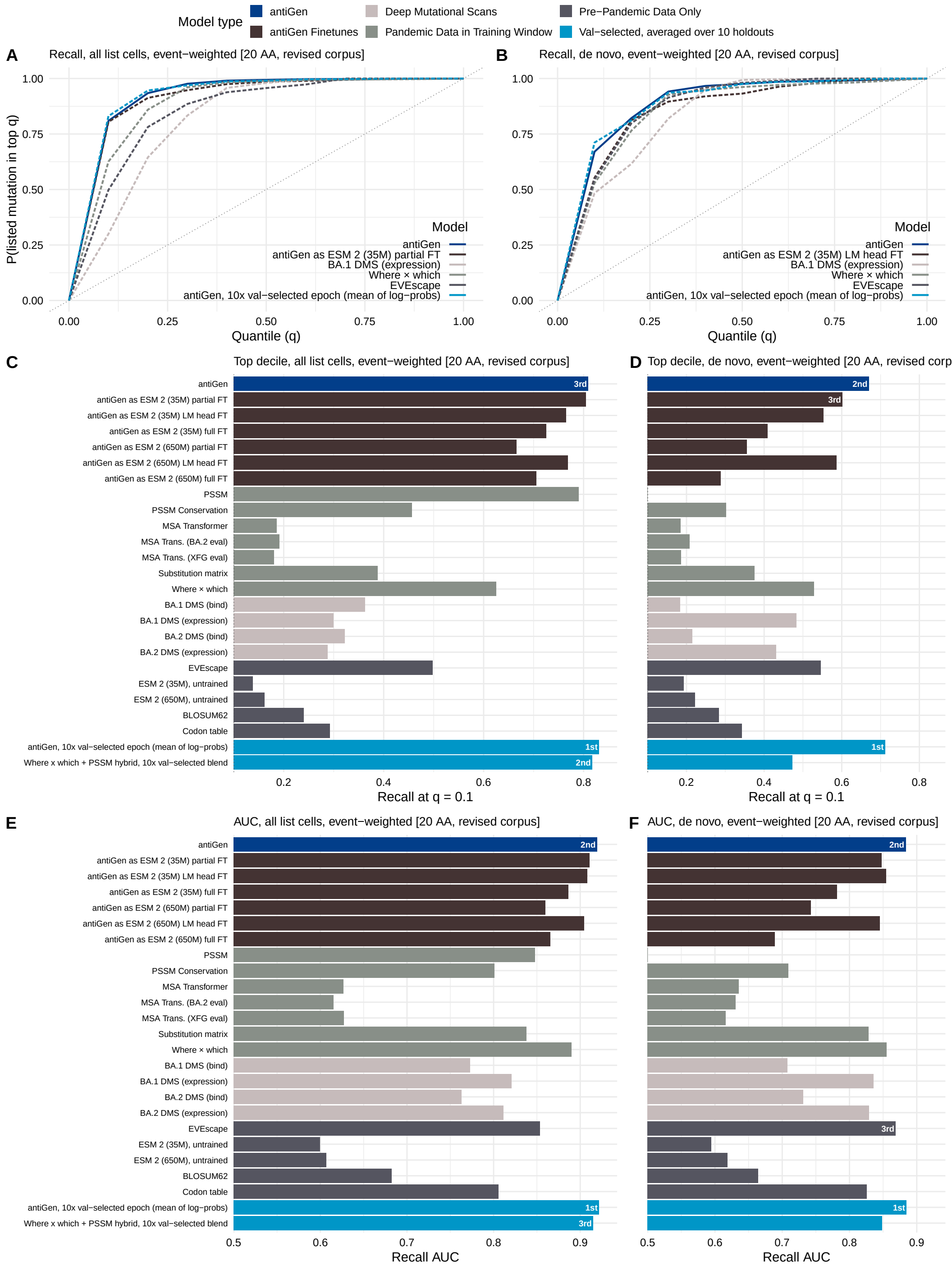
Model type
antiGen
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selected, averaged over 10 holdouts
A
Recall, all list cells, event−weighted [20 AA, revised corpus]
P(listed mutation in top q)
Quantile (q)
Model
antiGen
antiGen as ESM 2 (35M) partial FT
BA.1 DMS (expression)
Where × which
EVEscape
antiGen, 10x val−selected epoch (mean of log−probs)
B
Recall, de novo, event−weighted [20 AA, revised corpus]
Quantile (q)
Model
antiGen
antiGen as ESM 2 (35M) LM head FT
BA.1 DMS (expression)
Where × which
EVEscape
antiGen, 10x val−selected epoch (mean of log−probs)
C
Top decile, all list cells, event−weighted [20 AA, revised corpus]
antiGen
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
Substitution matrix
Where × which
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
antiGen, 10x val−selected epoch (mean of log−probs)
Where x which + PSSM hybrid, 10x val−selected blend
3rd
1st
2nd
Recall at q = 0.1
D
Top decile, de novo, event−weighted [20 AA, revised corp
2nd
3rd
1st
Recall at q = 0.1
E
AUC, all list cells, event−weighted [20 AA, revised corpus]
antiGen
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
Substitution matrix
Where × which
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
antiGen, 10x val−selected epoch (mean of log−probs)
Where x which + PSSM hybrid, 10x val−selected blend
2nd
1st
3rd
Recall AUC
F
AUC, de novo, event−weighted [20 AA, revised corpus]
2nd
3rd
1st
Recall AUC